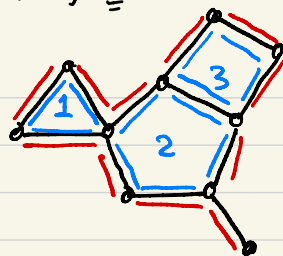


Theorem: In any planar graph $e \leq 3v-6$ and $f \leq 2v-4$.

Proof: Number faces $1, 2, \dots, f$

$m_i = \# \text{ edges on face } i$



$m_1 = 3$
 $m_2 = 5$
 $m_3 = 4$
 $m_4 = 11$

Count sum of m_i in two different ways

$$(1) m_1 + m_2 + \dots + m_f = \sum_{i=1}^f m_i \geq \sum_{i=1}^f 3 = 3f$$

each m_i is at least 3



$$(2) m_1 + m_2 + \dots + m_f = \sum_{i=1}^f m_i \leq 2e$$

because each edge is counted for at most two faces.



$$3f \stackrel{(1)}{\leq} \sum_{i=1}^f m_i \stackrel{(2)}{\leq} 2e$$

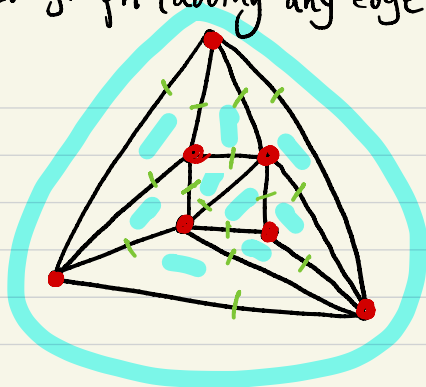
$$\Rightarrow 3f \leq 2e \Rightarrow f \leq \frac{2}{3}e \quad (**)$$

$$(*) \quad e \leq v + f - 2 \leq v + \frac{2}{3}e - 2$$

$$e \leq v + \frac{2}{3}e - 2 \xrightarrow{\times 3} 3e \leq 3v + 2e - 6 \rightarrow e \leq 3v - 6 \quad \checkmark$$

$$(**) \quad f \leq \frac{2}{3}e \leq \frac{2}{3}(3v - 6) = 2v - 4 \Rightarrow f \leq 2v - 4 \quad \checkmark$$

maximal planar graph (adding any edge creates a crossing)



called Triangulations
(every face formed by 3 edges)

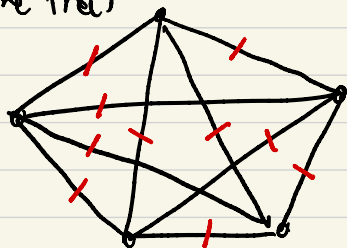
$$v = 7$$

$$f = 10 = 2v - 4 = 2 \times 7 - 4 = 10$$

$$e = 15 = 3v - 6 = 3 \times 7 - 6 = 15$$

Why K_5 is not planar?

Prove that



Proof by contradiction:

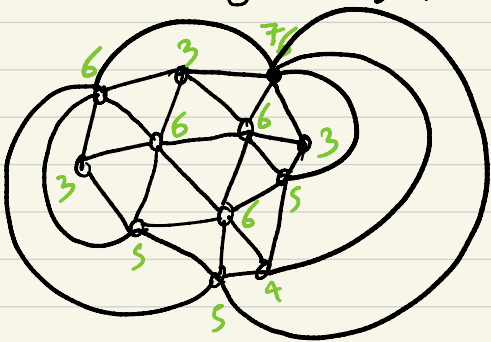
assume K_5 is planar, $v = 5$

$$\Rightarrow e \leq 3v - 6 = 3 \times 5 - 6 = 9$$

this contradicts the fact that

K_5 has 10 edges. ■

Theorem: any planar graph has a vertex of degree at most 5.



Proof:

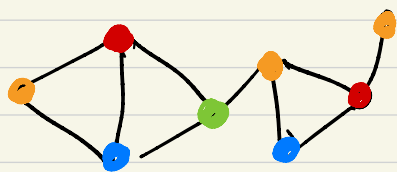
$$\sum_{u \in V} d(u) = 2e \leq 2(3v - 6) = 6v - 12$$

$$\text{average degree} = \frac{\sum_{u \in V} d(u)}{\# \text{ vertices}} \leq \frac{6v - 12}{v} = 6 - \frac{12}{v} < 6$$

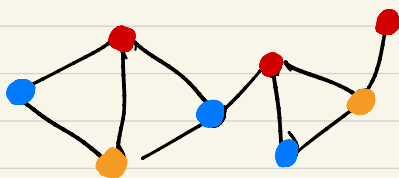
Since the avg. deg. < 6 there must be a vertex p of degree < 6 .

Since the degrees are integers then the degree of p is ≤ 5 . ■

Graph Coloring (vertex coloring / proper coloring) : color the vertices of a graph s.t. any two adjacent vertices have different colors.



4-coloring

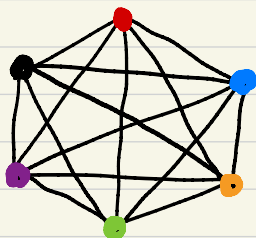


3-coloring

Is 2-coloring possible? no

Main Question: color the graph with min # of colors (NP-hard)

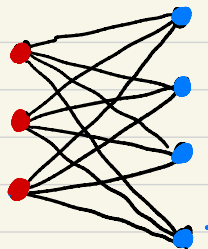
How about K_n ?



K_6

we need n colors

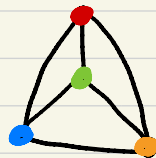
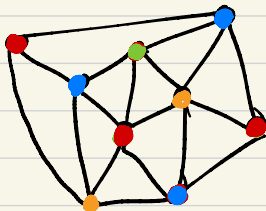
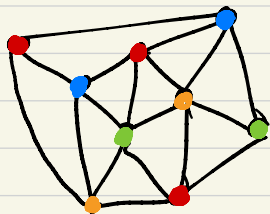
How about $K_{n,m}$?



$K_{3,4}$

2 colors suffice

How about planar graphs?

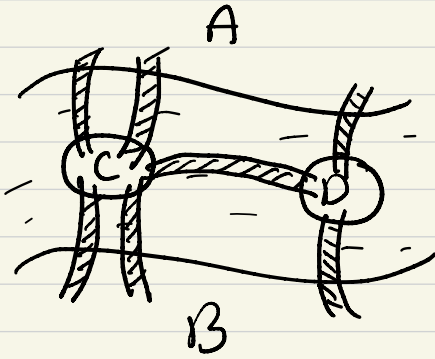


K_4

Can we color this with 3 colors? we need 4 colors here

Theorem (Four Color Thm.) : any planar graph can be 4-colored.

Theorem: Any planar graph can be colored by 6 colors.



Is it possible to start somewhere and walk over every bridge exactly once.

